

Final exam

Exam Instructions

- The exam is designed to take **75 minutes**.
- You may refer only to your **2-page cheat sheet** during the exam. All other resources, including personal notes, online content, and AI tools, are strictly prohibited.
- After the exam, please **submit your cheat sheet and scratch paper** via Canvas: (Canvas->Assignments->Final->[Supporting Work Upload](#)).
- On your scratch paper, **clearly explain the process** used to derive each answer. Answers without supporting explanations or derivations will not receive full credit.
- Show all **intermediate steps and derivations** clearly. Unreadable or missing supporting work will not be graded.
- You must complete the exam independently. Collaboration with others in any form is strictly prohibited.
- Do not share or discuss exam questions or content with anyone during or after the exam period.
- Any form of cheating or plagiarism will result in a zero on the exam and may lead to further disciplinary action.
- Your work must be entirely your own. Any content that closely resembles material generated by AI tools or submitted by another student, *even if reworded*, will be treated as a serious academic integrity violation.

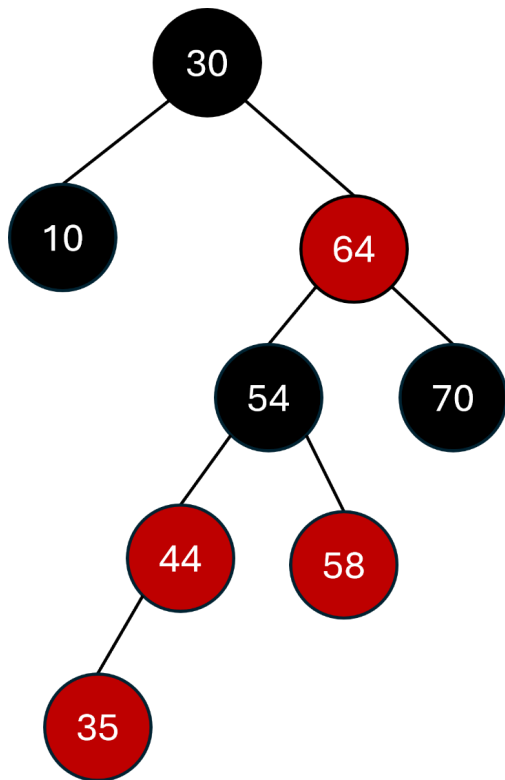
1

Multiple Choice 1 point

I pledge on my honor that I have not given or received any unauthorized assistance on this assignment/examination.

- ☐ I disagree.
- ☐ I agree.

The following tree is shown immediately after inserting the value 35 into a red-black tree.



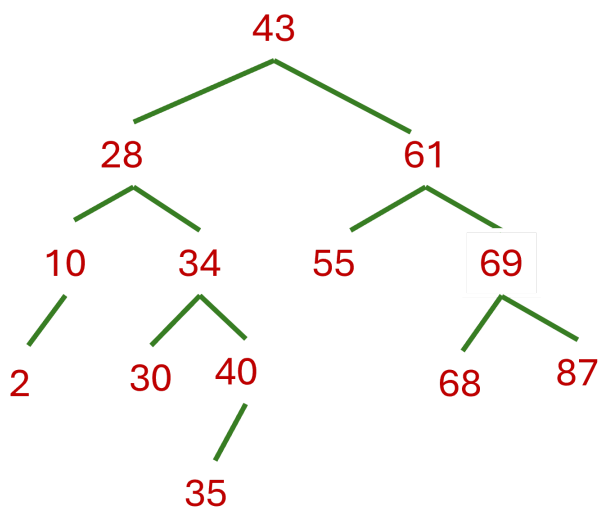
Select all statements that are correct:

- ☐ The inserted node's **uncle is red**.
- ☐ The tree violates the **red property**.
- ☐ The tree is **not balanced** because the difference between the left and right subtree heights is greater than 1.
- ☐ The tree violates the **black-height property**.
- ☐ New nodes are **always inserted as red**, before the rebalancing process begins.

Instructions

Consider the following AVL tree for the next two questions.

AVL Tree



3

Multiple Answer 3 points

A new node with value 90 is being inserted into this AVL tree. Before any rebalancing occurs, identify all nodes that become unbalanced as a result of the insertion.

- ☐ 43
- ☐ 69
- ☐ 34
- ☐ 28
- ☐ 61
- ☐ 87
- ☐ 55

4

Essay 5 points

On your scratch paper, show the sequence of rotations required to rebalance the tree after inserting the node with value 90. Then, write the level-order array representation of the final balanced AVL tree below.

Grading Criteria:

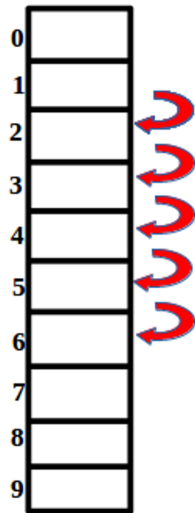
- Correct final answer (array representation): 2 points
- Correct and clear demonstration of balancing operations (partial credit available): 3 points

Instructions

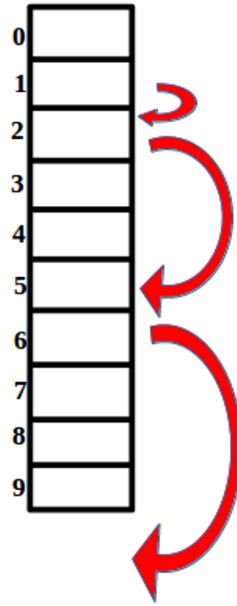
You are given visualizations related to open addressing hash tables using different probing strategies.

Hashing

Method 1



Method 2



$m=10$, Key={10, 20, 30, 40, 50}, $h'(k) = k \bmod m$

Key	Initial Hash $h'(k)$	Behavior A	Behavior B
10	$10 \bmod 10 = 0$	Inserts at index 0	Inserts at index 0
20	$20 \bmod 10 = 0$	$0 \rightarrow 1$	$0 \rightarrow 1$
30	$30 \bmod 10 = 0$	$0 \rightarrow 1 \rightarrow 2$	$0 \rightarrow 1 \rightarrow 4$
40	$40 \bmod 10 = 0$	$0 \rightarrow 1 \rightarrow 2 \rightarrow 3$	$0 \rightarrow 1 \rightarrow 4 \rightarrow 9$
50	$50 \bmod 10 = 0$	$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$	$0 \rightarrow 1 \rightarrow 4 \rightarrow 9 \rightarrow 6$
60	$60 \bmod 10 = 0$	$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$	$0 \rightarrow 1 \rightarrow 4 \rightarrow 9 \rightarrow 6 \rightarrow 5$
70	$70 \bmod 10 = 0$	$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$	$0 \rightarrow 1 \rightarrow 4 \rightarrow 9 \rightarrow 6 \rightarrow 5 \rightarrow 9 \rightarrow 4 \rightarrow 1 \rightarrow 0$

5

Multiple Answer 4 points

Select all correct statements based on the behaviors of Method 1, Method 2, Behavior A, Behavior B as depicted in the images and table above.

- ☐ Chaining does not suffer from these behaviors.
- ☐ Behavior B forms longer contiguous clusters compared to Behavior A.
- ☐ These behaviors result in a load factor greater than 1.
- ☐ Even a well-designed hash functions can lead to these behaviors depending on how the input data is distributed.
- ☐ These behaviors arise when multiple keys hash to the same index and follow identical probing sequences.

6

Multiple Choice 2 points

Consider the following hashing method:

$$h(k, i) = (h_1(k) + i \cdot h_2(k)) \bmod m$$

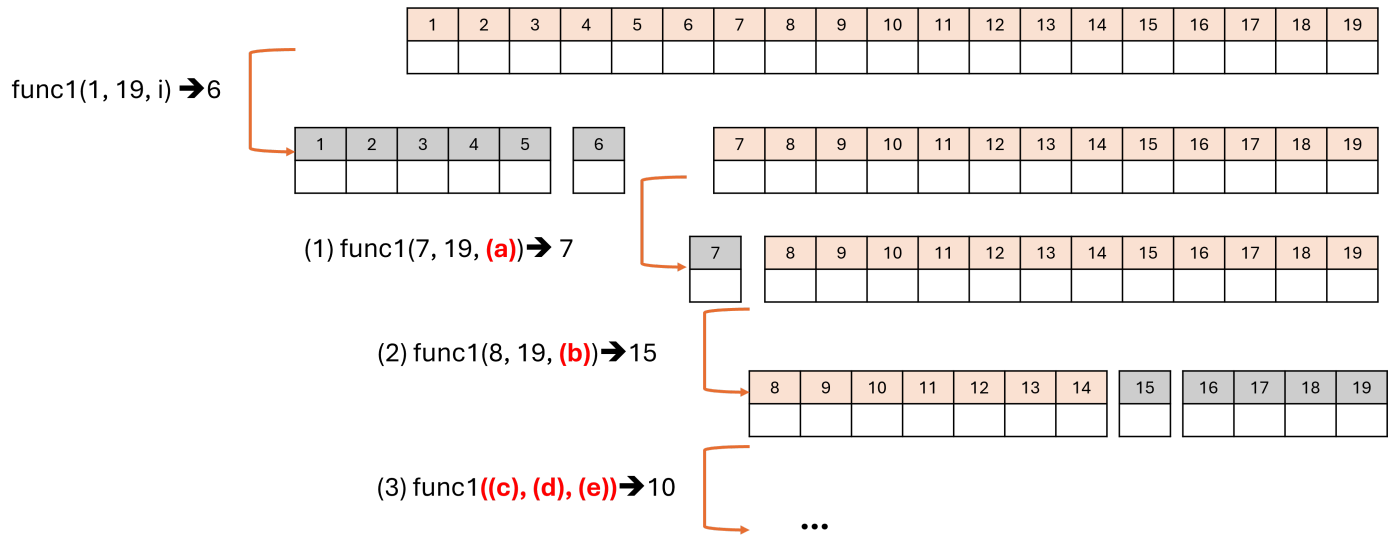
What is the main idea behind this approach?

- ☐ You randomly choose h_1 and h_2 each time a hash table is generated.
- ☐ Even if two keys have the same initial hash values, their probing sequences will differ.
- ☐ The modulo operation ensures that all keys hash to unique indices.
- ☐ There is a very high chance that Behavior B will still occur.
- ☐ Both h_1 and h_2 are precomputed based on known keys to prevent collisions.

Instructions

The following visualization demonstrates how func1 identifies the i-th order statistic. The top row displays element indices, while actual values are intentionally omitted for simplicity.

Order Statistic



7 Fill in the Blank 5 points

(1) func1(7, 19,)

(2) func1(8, 19,)

(3) func1(, ,)

8 True or False 1 point

This algorithm follows a divide-and-conquer strategy.

- ☐ True
- ☐ False

9 True or False 1 point

The expected time complexity of this algorithm is $O(n)$.

- ☐ True
- ☐ False

10 True or False 1 point

The Median-of-medians algorithm guarantees to choose the median value for the pivot element at every step.

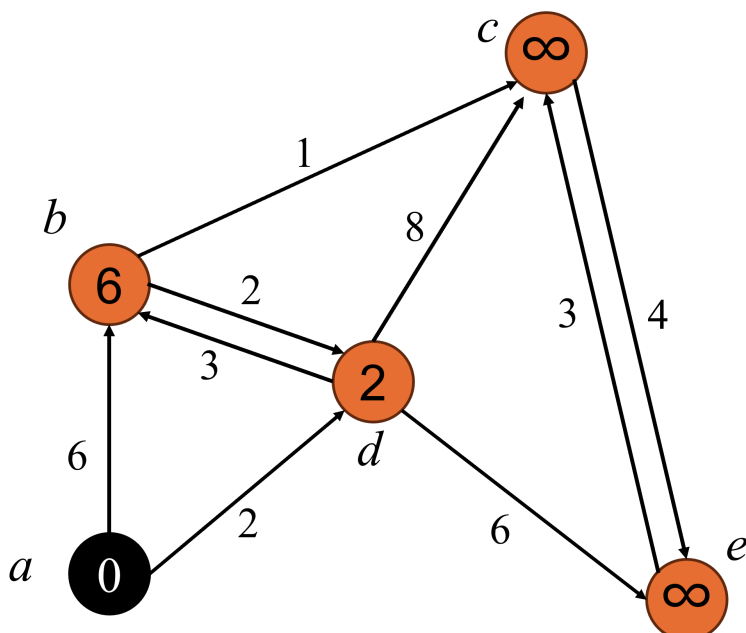
- ☐ True
- ☐ False

Instructions

Consider the following pseudocode for Dijkstra's algorithm and the graph. A black-colored vertex indicates that it is in the set S .

Dijkstra

1. $Q \leftarrow V[G]$ and $S \leftarrow \emptyset$
2. for each vertex $u \in Q$
3. do $d[u] \leftarrow \infty$ and $\pi[u] \leftarrow \text{NIL}$
4. $d[s] \leftarrow 0$
5. $\pi[s] \leftarrow \text{NIL}$
6. while $Q \neq \emptyset$
7. do $u \leftarrow \text{Extract-Min}(Q)$
 $S \leftarrow S \cup \{u\}$
8. for each $v \in \text{adj}[u]$
9. do if $v \in Q$ and $(d[u] + w(u,v) < d[v])$
10. then $\pi[v] \leftarrow u$
11. $d[v] \leftarrow d[u] + w(u,v)$
12. decrease_Key($Q, v, d[v]$)



11 Multiple Answer 4 points

Select all correct statements that are generally true about Dijkstra's algorithm.

- ☐ In general, there may be multiple shortest paths between two vertices in a graph, but the shortest path length $\delta(s, u)$ is unique.
- ☐ Dijkstra's algorithm works on both weighted and unweighted graphs.
- ☐ Dijkstra's algorithm does not always guarantee an optimal solution because it is a greedy algorithm.
- ☐ Dijkstra's algorithm waits until all vertices are visited before choosing the best option.
- ☐ When a vertex u is extracted from the priority queue Q , its distance value satisfies $d[u] = \delta(s, u)$.

12 Multiple Choice 2 points

Which vertex is selected as u in the next step?

- ☐ c
- ☐ d
- ☐ b
- ☐ e
- ☐ a

13 Multiple Answer 3 points

Based on your answer to the previous question (#12), where u is the vertex you selected:

Which vertices are considered for relaxation? (Select all that apply)

- ☐ d
- ☐ c
- ☐ a
- ☐ e
- ☐ b

14

Fill in the Blank 4 points

Complete the table by filling in the blanks after all relaxations from u have been processed, but before the next Extract-Min operation.
(Use your selected vertex u from Question #12.)

Use **NIL** to indicate a null predecessor and **INF** to represent ∞ (infinity).

	d	π
a	0	NIL
b		
c		
d		
e		

15

Multiple Answer 4 points

Select all correct statements about graph algorithms:

- ☐ Topological sort does not work if the graph contains one or more strongly connected components.
- ☐ Cycles in a graph can be detected during DFS by identifying back edges.
- ☐ DFS always finds the shortest path in an unweighted graph.
- ☐ A minimum spanning tree does not contain any articulation points.
- ☐ Minimum spanning trees are typically computed using DFS traversal.

DP

Consider the following table M, used for solving matrix chain multiplication using dynamic programming.

Note: Please note that the table is partially filled: cells that appear empty or contain "..." may represent either missing values or values that are intentionally omitted.

	1	2	3	4	5	6
1	0	120				
2		...	48			
3			...	...		
4				...	...	
5					...	...
6						...

M

16 Multiple Choice 2 points

What does $M[i,j]$ represent in the context of matrix chain multiplication?

- ☐ The product $p[i-1] \cdot p[k] \cdot p[j]$ used in cost calculation.
- ☐ The optimal cost of multiplying the chain from A_i to A_j .
- ☐ The resulting matrix from multiplying A_i to A_j .
- ☐ The number of scalar multiplications required to compute $A_i \times A_j$.

17 Multiple Answer 4 points

What is correct or what can you infer from this table? (Select all correct statements.)

- ☐ Diagonal entries $M[i,i]$ serve as the base cases and are always 0.
- ☐ The final solution is located at $M[6,6]$.
- ☐ $M[1,2] = 120$ indicates that 120 scalar multiplications are required to compute $A_1 \times A_2$.
- ☐ There are six matrices to multiply.
- ☐ The tabulation method computes the optimal parenthesization by recursively dividing the matrix chain.

18

Numeric 1 point

For the matrix chain $A_1 A_2 A_3 A_4 A_5 A_6$, given the dimension array $P = \langle 2, 3, 4, 5, 2, 3, 4 \rangle$, compute $M[3,5]$.

On your scratch paper, clearly show:

- Base cases
- Relevant subproblems
- Intermediate steps used to compute $M[3,5]$

Grading criteria

Correct final answer: 1 point

Base case shown: 1 point (evaluated manually)

Relevant subproblems identified: 1 point (evaluated manually)

Intermediate steps for computing $M[3,5]$: 2 points (evaluated manually)

Answer